MILITARY SPECIFICATION

DRILLING SYSTEM, WATER WELL, ROTARY, MINIMUM 1500 FEET CAPACITY AIR TRANSPORTABLE

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This purchase description covers a rotary water well drilling system capable of drilling to a 1,500 foot depth. The components of the well drilling system are the drilling machine (rig), support vehicle, and water well pumping system. The drilling system also includes the tools, supplies, and accessories necessary to develop a water well producing 50 GPM. The drilling machine and support vehicle shall be mounted on diesel-engine-driven multi-axle trucks suitable for on road and off road transport. The water well pumping system shall be trailer mounted. Each component shall be transportable in a C-130, C-141, C-17 and C-5 aircraft without removal of major assemblies.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Issues of documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and the supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

KKK-T-2111	Trucks and Truck Tractors: Commercial, Diesel Engine Driven, 34,500 to 66,000 Pounds GVW, 6 X 6
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-8625	Anodic Coating, For Aluminum and Aluminum Alloys
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, For Aerospace Applications

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/LZMD, 226 Cochran St, Robins AFB, GA. 31098-1622, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3820

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STANDARDS

Federal

FED-STD-297	Rustproofing of Commercial (Non-tactical) Vehicles Colors
FED-STD-595	Colors

Military

MIL-STD-129	**Marking for Shipment and Storage**
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-831	Test reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Non-tactical Wheeled Vehicles Treatment, Painting Identification Marking and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities
MIL-STD-1474	Noise Limits
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MS51335F	Pintle Assembly, Towing, 18,000 lbs Capacity, Manual Release
MS51336	Lunette - Coupler, Drawbar, Ring
MS52131	Connector, Plug, Electrical Intervehicle Power Cable

(Unless otherwise indicated, copies of Federal and Military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave, Philidelphia, PA. 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Goverment documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shalllbe those in effect on the date of the solicitation.

DRAWINGS

U.S. AIR FORCE

8838302A	Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and support Equipment

(Copies of this drawing shall be obtained from WR-ALC/LVRG, Robins AFB, GA 31098-5345.)

(Unless otherwise indicated, copies of the federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents specified in the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to SAE, Incorporated, 400 Commonwealth Drive, Warrendale, PA. 15096.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, see 6.2 part c, the contractor shall furnish all or any portion of the well drilling system for first article inspection and approval (see 4.4 and 6.2 part c).

3.2 Description.

3.2.1 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperatures that are of such nature or are located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. Ladders, steps and handrails shall be provided so entrance and exit are unhampered and nonhazardous. All brake pads and braking materials shall be made from asbestos free components.

3.2.2 Maintainability. The well drilling system components shall be designed such that components are readily accessible for repair or replacement, with minimum removal or disturbance of adjacent parts or components using general purpose tools. Operating parts, accessories and drain outlets shall be easily accessible for regular maintenance service. The vehicle(s) shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating

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trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime of no greater than one hour.

g. Maximum corrective maintenance downtime of no greater than four hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

The vehicle(s) shall be designed so that one mechanic, using hand tools, can perform the following maintenance operations within the time specified:

(1) Remove, replace and adjust all engine driven belts - 3/4 hour.

(2) Remove and replace alternator - 1/2 hour.

(3) Remove and replace all filters, screens and strainers in all hydraulic systems - one hour.

(4) Remove and replace engine coolant hoses - 1/2 hour.

(5) Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase - one hour.

(6) Remove and replace fuel filter elements - 1/2 hour.

(7) Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1/2 hour.

(8) Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission, if applicable, - one hour.

(9) Remove and replace starter - one hour.

(10) Remove or install engine/transmission package - two mechanics, six hours.

(11) Rebuild mud pump, complete - 2 operators, 8 hours.

(12) Repair mud pump, sleeves, pistons, and packings - 2 operators 1 hour.

(13) Adjust main clutch - 2 mechanics, 1 hour.

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(14) Replace and adjust hoist drum clutch - 2 mechanics, 2 hours.

(15) Adjust auxiliary drum clutch, 2 mechanics, 1 hour.

(16) Adjust sand reel clutch - 2 mechanics, 1 hour.

(17) Remove and replace all hoist cables - 2 mechanics, 2 hours.

(18) Adjust all hoist drum brakes - 1 mechanic, 1/2 hour each brake.

(19) Replace mud pump oil stop and fluid packing (seals) 1 mechanic, 1 hour.

(20) Dismantle and rebuild water injection pump - 1 hour.

3.2.3 Operational environment. The well drilling machine and support vehicle shall be capable of traveling in temperatures from -20 °F to 120 °F at sea level and in temperatures from 0 °F to 120 °F when traveling at altitudes of up to 8,000 feet above sea level.

3.3 Preproduction. Unless otherwise specified, see 6.2, the contractor shall furnish the specified components of the well drilling system for preproduction inspection in accordance with Sections 4 and 6.

3.4 Commerciality. The well drilling system components, accessories, attachments and supplies shall be the manufacturer's standard commercial products **except as modified to meet the requirements of this document which have been** manufactured and sold to the general public at least one year prior to the opening date of the solicitation. Offerers shall submit evidence to establish commerciality of their system components.

3.5 System chassis. The system chassis shall be type I, class B, diesel engine driven, multi-axle, six wheel drive (6 X 6) commercial truck chassis with front mounted winch, and operator cab in accordance with KKK-T-2111. The diesel engines shall be capable of operating all equipment permanently mounted on the chassis. The support vehicle shall be mounted on a chassis conforming to KKK-T-2111, of the same manufacture to ensure maximum commonality of parts and on-site support. The chassis shall be equipped with a NATO type slave starting system in conformance with MS52131.

3.5.1 Chassis tires. Each chassis shall be equipped with the manufacturers commercial standard tires meeting all the performance requirements specified herein.

3.5.2 Weights and dimensions. The overall dimensions of the vehicles shall be no more than:

Length	420 inches
Width	96 inches
Height	102 inches

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The axle loads of the vehicles shall be no more than:

Single steering axle	13,000 pounds
Dual rear axle	26,000 pounds
Gross vehicle weight (dual rear axle)	39,000 pounds

The maximum dimensions and drive-on weight of the vehicle must meet the above air transportability requirements in accordance with MIL-STD-1791.

3.5.3 Human factors engineering. The drilling system shall conform to Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given but not limited to, 4 (General Requirements), 5.3 (Audio Displays), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for maintainability) and 5.13 (Hazards and Safety) of MIL-STD-1472 as applicable.

3.5.4 Chassis electrical system. The vehicles electrical system may be either a standard commercial 12-volt or 24-volt system. If a 12-volt system is offered, all wiring shall be upgraded to reduce the risk of electrical overloading and appropriate size fuses shall be installed to protect the vehicle and all operating personnel. The vehicles electrical system shall meet all power and load requirements as specified herein and shall comply with all applicable standards, codes and regulations. The vehicles electrical system shall be compatible with each other and conform to all applicable specifications and standards sited within this document.

3.6 Well drilling machine. When specified, see 6.2, a water well drilling machine shall be furnished. The water well drilling machine shall be capable of drilling to a depth of 1,500 feet through consolidated and unconsolidated soil formations of all types. The drilling machine shall consist of a derrick, hoisting assemblies, drilling machinery, and mud pump driven by a diesel engine through an appropriate power take-off mechanism, all installed on a multi-axle truck chassis. The transporter truck shall be a Type I, Class B truck chassis with cab in accordance with Federal Specification KKK-T-2111. The chassis shall be powered by a diesel engine with not less than 210 gross horsepower and be capable of operating all the required drilling equipment to a minimum depth of 1,500 feet. The well drilling machine shall be air transportable in C-130, C-141, C-17 and C-5 aircraft. The drilling machinery shall be installed on the drill frame to withstand all operational and transport forces specified herein without misalignment, malfunction, damage, or permanent deformation.

3.6.1 Performance. The well drilling machine shall be capable of drilling through soil and rock formations as well as setting 20 feet lengths of casing in accordance with the minimum performance requirements of drilling 1,500 feet deep with a bit size of 12-1/4 inch and setting 8-5/8 inch outside diameter casing. The drilling machine shall meet this minimum performance requirement when using either mud or air rotary techniques. In addition, the drilling machine shall be capable of drilling a 6-1/2 inch diameter hole through hard rock to a depth of 1500 feet using down-the-hole percussion technique in 100 hours or less. The drilling machine shall have both air and foam drilling capability. (Air for air or foam operations will be furnished from an external trailer mounted air compressor. Unless purchased as part of the drilling system

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and requested for first article testing, the manufacturer shall furnish a compressor capable of delivering a minimum of 900 CFM at 350 psi for the test when first article testing is required.) The well drilling system shall drill a 12-1/4 inch hole to a depth of 1,000 feet in 60 hours, continue with an 8 inch hole to a depth of 1,500 feet in 100 hours. A production water well shall be completed within 180 consecutive hours as specified in 4.4.4.

3.7 Support vehicle. When specified, see 6.2, a support vehicle shall be furnished. The water well drilling support vehicle shall be capable of carrying auxiliary equipment, tools and accessories required for well drilling operations, including water, fuel and drill pipe. The support vehicle shall be mounted on a Type I, class B truck chassis with cab in accordance with Federal Specifications KKK-T-2111. The support vehicle shall be powered with a diesel engine with a minimum of 210 gross horsepower and be capable of operating all equipment permanently mounted on the support vehicle. When the support vehicle is purchased in conjunction with the well drilling machine the support vehicle shall be mounted on the same model chassis and type of manufacturer to ensure maximum commonality of parts and on site support. The support vehicle shall be air transportable on C-130, C-141, C-17 and C-5 aircraft, (see 3.5.2 and 3.20).

3.8 Well drilling machine. Components of the well drilling machine shall be, but not limited to, the following:

a. Draw works.
b. Rotary table.
c. Sand reel.
d. Pull down/Hold back.
e. Mast and crown block.
f. Transfer case.
g. Hydraulic leveling jacks.
h. Controls.
i. Mud pumps.
j. Air line lubricator.
k. Breakout tongs.
l. Water injection system.
m. Crown block lubricator.

3.8.1 Draw works. The draw works shall be double drum equipped with heavy duty clutch and brake band. The hoisting capacity shall be rated at 15,000 pounds minimum single line pull on bare drum. The draw works shall have a minimum working hook load of 22,500 pounds with traveling block located at the top of the mast. Each drum shall be equipped with 5/8 inch nonrotating 18 X 7 wire rope for 2-part line operation. The draw works shall be capable of setting 8 inch steel casing up to 700 foot depth.

3.8.1.1 Sand reel. The drilling machine shall have a sand reel. The sand reel shall have a heavy duty clutch and brake band. Line pull on bare drum shall be 4,000 pounds minimum. The drum shall be equipped with a minimum of 2,000 feet of 3/8 inch non-rotating cable. The cable shall have a clevis mounted on the free end. A single sheave snatch block with hook shall be furnished for sand reel operations.

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3.8.2 Rotary table. The rotary table shall be mounted on a hydraulically operated sliding frame which allows for the placement of 14 inch OD casing. Table movement shall be controlled by the operator at the operators station. The kelly shall be 4-1/4 inch minimum diameter and fluted with keys. The machine shall have a heavy duty swivel with a minimum 17 foot length and a minimum 2 inch diameter swivel hose, a kelly and mud bucket. Minimum operational speed range of the table shall be from 0 to 200 RPM at a minimum torque of 135,000 inch pounds below 40 RPM. The table shall be equipped with a heavy duty dust deflector, including a positive seal wiper, to prevent sand, cuttings, dust or water from being blown in the operator's face when drilling with air. The dust cover shall allow for installation and removal from the drill table without removal of the drill pipe from the hole. A rigid stop shall be positioned on either the top or bottom of the drill platform to provide an appropriate amount of leverage on the handle of a pipe wrench for power breakout. A 48 inch, high quality, steel pipe wrench shall be supplied to complete the breakout assembly.

3.8.3 Pull down with holdback. The drilling machine shall have a hydraulic, chain or wire rope feed pulldown/holdback system. The system supplied shall have a minimum of 13,000 pounds of downward pull and a minimum of 30,000 pounds of holdback force. When hydraulic cylinders are supplied, the feed system shall have two speeds. The 'slow' speed shall have infinitely variable feed rates from a minimum of 11 feet per minute (fpm) upwards to 18 fpm downwards. The 'fast' feed rate shall be variable from 53 fpm minimum upwards to 89 fpm downwards. The pulldown force shall automatically maintain bit weight when set by the operator over the full length of the drill rod within the derrick assembly. When the system is chain or wire rope controlled, the swivel yoke assembly shall provide continuous feed over the full length of the kelly. The feed rate shall operate smoothly and shall provide variable rates from 0 to 12 fpm. If a top mounted motor is provided, a removable brush cover shall be provided to protect the components from damage during travel through undomesticated terrain.

3.8.4 Mast and crown block. The mast shall provide a working space sufficient to handle 20 foot casing lengths above the rotary table. Design of the mast shall permit retraction of the swivel assembly to allow casing setting with no obstructions within the mast assembly. The crown block and mast assembly shall be rated to a minimum capacity of 45,000 pounds. The mast shall be equipped with 6 each shock proof quartz iodine lights or equal. Four lights shall be positioned atop the mast, one at each corner to allow coverage of the work area, and two positioned mid-mast to provide lighting for the draw works, mud pump, etc. An additional light shall be positioned under the rear of the rig to provide lighting directly on the hole and drill string. The switches for these lights shall be located at the operator's platform; the lights shall be powered by the truck's electrical system. The mast shall be equipped with quick acting breakout tongs. When not in use these tongs shall swing free to allow for placing drill collars or large casing. Foot steps shall be installed on the mast to allow for personnel to ascend and descend the mast while it is in the raised position.

3.8.5 Transfer case. The unit shall have a transfer case equipped with hydraulic pumps, and external clutches which deliver power to the mud pump. The case assembly shall be an oil bath type with roller chains and provisions for

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adjustment.

3.8.6 Hydraulic leveling jacks. The drilling machine shall have a minimum of 4 leveling jacks with automatic valving to prevent drift during set up and drilling. All jacks shall be controlled from the drill operator's station, each operated independently with speed control. Each jack shall have a removable base plate and security chain.

3.8.7 Drilling station. The drilling machine shall be equipped with a drilling station containing all controls and instruments specified herein. The operator's platform shall be so located as to assure the operator of a clear view of the drill pipe at its point of entry into the well. The operator's platform shall be either swing or bolt-on type. All controls shall be accessible to the operator in the normal operating position. Instrument and gages shall be visible to the operator in his normal operating position. When not in use, the instruments and gages shall be protected by a weatherproof covering. Each instrument, gage, and control shall be provided with permanently affixed identification markings. Good human engineering considerations shall be used to provide positive and responsive control over the hydraulic and mechanical functions. Where practical, control knobs shall be configured for ease of identification by touch and location. Control knobs shall be consistent with the related equipments direction of movement. All pressure gauges shall be liquid filled to reduce needle flutter. The drilling station shall be equipped with, but not limited to the following instruments and controls:

a. Lubricating oil pressure indicator.
b. Cooling liquid temperature indicator.
c. Tachometer.
d. Pulldown pressure.
e. Hold back pressure.
f. Mud pump pressure.
g. Battery/alternator indicator.
h. Manual speed control.
i. Emergency shut down device.
j. Normal shutdown device.
k. Illumination light switch.

3.8.8 Mud pump primary. The mud pump shall be complete with all hoses and fittings for drilling operations. The minimum pump size acceptable is 5-1/2 X 8 with a minimum capacity of 260 GPM at 338 psi. The pump shall be equipped with a 5 inch suction and a 2-1/2 inch discharge. The mud pump shall be of a self-priming gear or chain driven, or direct hydraulic powered type. Power shall be supplied through a torque tube assembly. Packing glands and rod oiler shall be provided to reduce excessive piston rod wear under harsh environmental conditions.

3.8.8.1 Mud pump secondary. The secondary screw pump capable of pumping 15 GPM at 150 PSI at 930 RPM minimum shall be furnished. It shall contain all hoses and fittings to mix drilling fluids and grout during drilling operations. This pump shall be located on the drilling machine or support vehicle.

3.8.9 Air line lubricator. The air line lubricator shall be of sufficient

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size and capacity to meet all performance requirements specified herein.

3.8.10 Water injection system. The water injection system shall consist of a triplex piston pump driven by a hydraulic motor. The pump shall be capable of delivering 25 GPM at 700 psi. The pump output shall be controlled by a flow regulator on the operators console. The injection system shall be supplied with a 30 foot suction hose and a strainer to prevent debris from entering the pump assembly and a 50 foot pressure hose with adjustable nozzle for machine wash down. Hose connections shall provide quick connect and disconnect capability on all pump discharge ports.

3.8.11 Crown block lubricator. Lubrication fittings (zerks) on the crown block shall be installed in such a manner to allow them to be lubricated from drill platform level

3.8.12 Illumination. The drilling mast shall be provided with a night lighting system (see 3.8.6). Electrical power for the lights shall be obtained from the electrical system of the truck engine and the voltage requirements of the lights shall be compatible with the truck electrical system. All night lighting shall be wired and fused independently from the truck engine. All wiring shall be protected against wearing surfaces, weathering, moisture and debris.

3.8.13 Tool Storage. The drilling machine shall be furnished with tool boxes and built in storage compartments to carry tools and accessories specified herein that are less than five feet in length. Boxes shall be constructed of sheet steel. All doors and lids shall be equipped with continuous hinges and latches. Latches shall keep doors and lids securely fastened when subjected to vibrations. In addition to the latches, each box and compartment shall be fitted with a staple hasp and padlock. Not less than six keys shall be furnished for the padlocks and any single key shall open all the padlocks. Tool boxes shall be securely fastened to the rig's bed in accessible locations. Built-in storage compartments shall be located under the rig's bed, but no portion of the compartment shall extend below the lowest extremity of the truck main frame. Compartment floors shall be decked with plywood. Plywood shall be exterior grade, 5 ply, and not less than 3/4 inch thick. Plywood shall not be spliced together to form a deck compartment. Compartment doors shall open in either a sideward or downward direction. Compartments and toolboxes shall be provided with drain holes in the bottom.

3.8.14 Mud Laboratory. The well drilling machine or frame assembly shall incorporate a storage cabinet with a portable Baroid mud laboratory Model 82100 or equal installed.

3.9 Well Drilling support Vehicle. When specified see 6.2, the contractor shall furnish a well drilling support vehicle for first article inspection.

3.9.1 Description. The well drilling support vehicle shall consist of a C-130, C-141, C-17 and C-5 air transportable 6 X 6 vehicle with winch. The vehicle shall be capable of carrying accessories and supplies required in support of drilling to a depth of 1,500 ft.; a 8,200 pound capacity remote operated crane; 200 gallon minimum fuel capacity; welding equipment gas and electrical; tool boxes; water and mud slurry pump; and a water tank with

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capacity of 1,000 gallons. The chassis shall be of the type meeting specifications of KKK-T-2111, see 6.2.1 and 3.5.

3.9.2 Components. Components of the support vehicle shall be, but not limited to, the following:

a. Waterbed.
b. Auxiliary fuel tank and transfer pump.
c. Storage Cabinets.
d. Tool boxes.
e. Night lighting.
f. Water heating.
g. Workbench.
h. Vise.
i. Crane.
j. Winch.
k. Water pump.
l. Welder/generator.
m. Towing provisions.
n. Tools.

3.9.2.1 Waterbed. The water carrying tank shall be a 1,000 gallon capacity unit of flat design, 20 feet long and 8 feet wide, constructed of 1/4 inch single span steel plate. The top deck shall have a superimposed load capacity of 30,000 pounds in any 4 ft X 4 ft area. The waterbed shall be designed with baffling lengthwise and crosswise, while still allowing the unobstructed movement from front fill to rear drain of sand, scale, and other residue which might otherwise accumulate on the bottom. The waterbed shall be of a flat bottom design with integrated wheel wells, mounted directly to chassis frame flange for lowest possible center of gravity. The wheel wells shall allow for maximum suspension travel and use of tire chains. The waterbed shall be mounted to the chassis with a coil spring system that allows the frame to flex independently of the waterbed. No welding to any part of the chassis frame is allowed. The waterbed shall have stake pockets and a chaining rail integrated into both sides and rear with stakes and retaining pins furnished. The waterbed shall have a water level indicator, drains, fills, and vents both fore and aft that allow for filling or draining in any position on uneven ground. All drains shall be equipped with 1/4 turn ball valves of brass and stainless steel. The waterbed must have a proven record of non-leakage under all types of drilling conditions for an extended period of time.

3.9.2.2 Auxiliary fuel tank and transfer pump. The vehicle shall have an auxiliary fuel tank with a capacity of 200 gallons minimum. The fuel tank shall be equipped with a fuel transfer pump, 50 feet of hose and fuel nozzle. The fuel tank shall have non-skid surface on top and cleanout capability with water and sediment drain. The hose and nozzle shall be stowed during transport. The tank shall be capable of accepting 13 GPM without fuel splashback during refueling operations. The fuel tank shall be of sufficient strength to provide protection of the chassis cab and its occupants from the sudden movement forward of materials and equipment carried on the water bed. The tank shall have a pressure relief type cap with capability of being removed by the operator wearing arctic mittens.

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3.9.2.3 Cabinets. The vehicle shall have two vertical cabinets of sufficient size to accommodate (1) a double set of welding/burning gas bottles and related tools; (2) a cabinet with a minimum of three removable shelves and 6 hooks of heavy duty size for support gear required when supporting drilling operations. All cabinets shall have drain holes in the bottom. The cabinet doors shall be equipped with continuous hinges and latches. Cabinet doors shall open in a sideward direction. A locking mechanism shall be provided for each cabinet. A total of six (6) keys shall be provided with any single key opening all of the cabinets and tool boxes.

3.9.2.4 Tool Boxes. The vehicle shall have a minimum of four heavy duty, horizontal tool boxes. The tool box shall have heavy duty continuous hinge down doors capable of being used as a work platform. The tool boxes shall have heavy duty latches requiring external lock and keys.

3.9.2.5 Night lighting. The support lighting shall be wired and fused independently of the chassis ignition system. The vehicle shall have two quartz iodine lights or equal, flood/worklights, debris guard protected, capable of lighting the deck and work area. The flood/worklights shall be capable of being manually elevated to a height of 4 ft above their normal travel position and have a swivel range of 360 degrees while in the elevated or travel position. All cabinets and tool boxes shall have lights with brush guard protection and individual switches.

3.9.2.6 Water heating. The waterbed shall be constructed with an integrated open-loop water heating manifold capable of being used as an in-transit storage chamber for the water pump hoses. A propane heating torch rated at 500,000 BTU shall be provided along with 25 feet of hose, a regulator and manifold mounting bracket.

3.9.2.7 Workbench. The vehicle shall have a heavy duty self-storing workbench located on the same side as the gas welding equipment.

3.9.2.8 Vise. The vehicle shall have a demountable 8 inch chain vise. The vise shall be located in close proximity to the workbench. The vise shall have permanent storage location for transit operations.

3.9.2.9 Crane. The crane shall be fully hydraulic with a minimum of 8,200 lb capacity, with 9 ft. boom at 3 ft. radius. The crane shall be hydraulically powered from the vehicle PTO system. The crane shall be hydraulically extendible **from 9 ft to 15 ft and manually extendible from 15 ft to 20 ft. The crane** shall have 360 degree continuous hydraulic power rotation in either direction, with an hydraulic power operated winch and 2 part hook block capability. All crane **functions shall be operated by a minimum 20 ft length hard wired and crane** stored remote control. The crane shall be of pedestal design mounted directly to the waterbed without outriggers. A self-storing operators platform shall be provided at the operator's console.

3.9.2.11 Winch. The vehicle shall be equipped with a minimum 20,000 lb capacity winch mounted at the front of the chassis. It shall be equipped with a minimum 150 feet of cable, tail chain, cable guide roller, heavy duty bumper

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and built-in tool boxes.

3.9.2.12 Water pump. The water pump shall be a minimum of 140 GPM capacity, reversible rotary gear design, self-priming, positive displacement, hydraulic driven complete with two 20 foot suction hoses and two 20 foot suction/fill hoses, one strainer and all with interchangeable quick couplings. The pump shall be able to handle slurries with high solids content. It shall be installed in such a way as to provide for pumping water into the waterbed, out of the waterbed, or from one area to another without going through the waterbed. The hoses shall be capable of being installed by one man.

3.9.2.13 Welder Generator. The welder/generator shall be a self-contained unit powered by an air cooled diesel engine. As a welder, it shall have a rating of 225 amps AC/200 amps DC @ 100% duty cycle. As a power plant, it shall have a rating of 4.5 KVA, 120/240 volts, 38/19 amps, 60 Hz AC (single phase). It shall be equipped with a 100 foot electrode and ground clamp, a rack for cable storage and canvas cover.

3.9.2.14 Tools. In addition to the requirements of para. 3.7, the equipment support vehicle must be able to carry on the flatbed and trailer (not part of **this specification) all items listed in para. 3.13 (Accessories). The** support vehicle shall be equipped with the following tools when specified in para. 6.2.

ITEM	DESCRIPTION	QUANTITY
1.	Heavy duty welding torch; welding range--2-200 CFH, acet & heating range--to 294,000 BTU/hr.	1
2.	Heavy duty cutting torch, 90 degree nozzle head	1
3.	No. 6 welding head	1
4.	No. 15 welding head	1
5.	No. 30 welding head	1
6.	No. 70 rosebud heating head	1
7.	No. 5 general purpose cutting nozzle	1
8.	Heavy duty oxygen regulator	1
9.	Heavy duty acetylene regulator	1
10.	1/4 in. X 50 ft. twin gas hose, b-b size	1
11.	Welding goggles, hard side	1
12.	Friction lighter	1
13.	Set, Operating instructions	1
14.	150 cu. ft. oxygen cylinders	2
15.	130 cu. ft. acetylene cylinders	2
16.	Cylinder T-wrench	1
17.	#7 mild steel gas rod, 1/8 in. X 36 in.	20 lbs.
18.	#1 high-tensile gas rod, 1/8 in. X 36 in.	20 lbs.
19.	1 lb. can brazing flux	1
20.	1 lb. can cast iron welding flux	1
21.	Std. set tip cleansers, No. 0-12, w/file	1
22.	Replacement flints for friction lighter	12
23.	Flat soapstone holder, 5 in. X 3/16 in. X 1/2 in.	1
24.	Flat soapstone, 5 in. X 3/16 in. X 1/2 in.	144
25.	1 lb. can cast iron brazing flux	1

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26.	Heavy duty welding helmet, under chin style, with flip-front, 4-1/4 in. X 2 in. glass	1
27.	Pair, all leather welding gloves, 13 in. long	1
28.	Steel chipping hammer	1
29.	Wire brush	1
30.	No. 7018 welding rod, 3/32 in. in sealed steel can	50 lbs.
31.	No. 6010 welding rod, 3/32 in. in sealed steel can	50 lbs.
32.	Clear glass spares, 4-1/4 in. X 2 in.	10
33.	No. 10 shade glass spares, 4-1/4 in. X 2 in.	10
34.	1/2 in. X 10 ft. high test towing chain with hook and shackle	1
35.	3/8 in. X 10 ft. binder chain for deck load	6
36.	3/8 in. chain binders	6
37.	25 ft. air hose with nozzle, tire gauge, and chassis air tool adapter	1
38.	50 ft. heavy duty 12 AWG 3 conductor extension cord	3
39.	1/2 in. electric hand drill, variable speed w/carrying case	1
40.	29 piece twist drill set	1
41.	13 Amp electric grinder with case	1
42.	7 in. 24 grit grinding disc	6
43.	110 volt 550 watt portable flood light w/tripod base and debris guard	2

3.9.3 Trailer mounted compressor. When specified, see 6.2, a self contained, trailer mounted screw type air compressor is required, capable of producing a minimum of 900 SCFM at 350 psig. The trailer shall have, as a minimum, a single front and rear axle, pneumatic tires and tongue with lunette eye hitch in accordance with MS51336. The trailer shall be capable of sustained towing at highway speeds (sustained 55 mph) without material degradation, excessive damage producing vibration or axle overheating and low speed towing over rough terrain. The entire unit must be capable of being transported and backed onto a C-130 aircraft by a prime mover without interfering with the crest of the ramp. Design and drawings must be furnished with the technical proposal for verification and approval by the USAF. The engine shall run below maximum load during normal operations. Technical data with specifications and manufacturer's expected standards along with the manufacturers continuous rating curve of the engine shall be provided for review by the USAF in the technical proposal. All operational components shall not exceed the maximum continuous rating curve of the engine. The compressor shall be capable of sustaining down-the-hole hammer drilling operations to a depth of 1500 feet with borehole sizes ranging from 6 inches to 8¼ inches. Drill pipe size will be 4½ inch OD with 2 inch ID. Annular uphole velocity shall be a minimum of 4000 ft/min. The compressor shall be of closed construction and shall operate at a continuous rate with the engine running at a constant rpm. The lubrication system oil shall be of POA or BOA base to prevent seal swelling and leaks to sustain optimum performance during operation. Technical data with specifications and engine manufacturers expected standards shall be provided with the technical proposal for evaluation by the USAF. The manufacturer shall also state the length and diameter of the screw compressor rotors at each stage and the normal operating rpm for verification of the maximum tip speed produced by the rotors during operation.

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3.10 Water well pumping system (WWPS). When specified see 6.2, a WWPS shall be furnished.

3.10.1 Description. The WWPS shall be trailer mounted, diesel electric powered, capable of pumping a 600 foot water head at 50 GPM. The WWPS shall be C-130, C-141, C-17 and C-5 air transportable without removal of major assemblies. The WWPS trailer shall be equipped with an air brake system compatible with the well drilling machine and support unit. The system shall be capable of being towed at highway speeds, and over terrain where roadways are nonexistent. The WWPS shall not exceed 28,200 U.S. pounds in the operating configuration and shall not exceed 13,000 pounds per axle in the air transportable configuration as per MIL-STD-1791.

3.10.2 Components. The system shall consist of, but not be limited to, the following:

a. Frame assembly.
b. Diesel electric power unit.
c. Hose reel assembly.
d. Submersible pump assembly.
e. Derrick assembly.
f. Flow meter assembly.
g. Night lighting.
h. 3000 gallon collapsible tank (1 ea.).

3.10.2.1 Frame assembly. The frame assembly shall be mounted on a single axle **or tandem axle with pintle hook for towing in accordance with MS51335F. The** frame assembly shall be capable of being towed by the well drilling or support vehicle at highway speeds. The frame assembly shall be capable of negotiating the C-130 aircraft loading ramp. Each frame assembly shall have a minimum of 3 jacks for stabilization of the WWPS during operation. The jacks shall have removable pads and shall be mechanically operated. The hose reel supports and derrick mount assembly shall be integral to the frame assembly.

3.10.2.2 Diesel electric power unit (DEPU). The DEPU shall have a KVA rating of 60 with 3 phase 4 wire 120-208-230 volt and 3 phase 460 volt rating. All on board service receptacles shall be no fault protected. The electrical system shall have a grounding point with a three-sectional 6 ft minimum length grounding rod with positive lock type connectors, with a 20 ft ground wire 10 AWG flexible stranded type. The DEPU shall have a fuel reservoir capable of operation for 12 hours without refueling.

3.10.2.2.1 Control panel. The DEPU shall have an integral control panel. The control panel shall be protected from environmental elements from all sides. The control panel shall include monitoring devices for engine water temperature, engine oil pressure, fuel level, etc. A/C monitoring meters, master circuit breaker, and individual controls for all electrical operated components of the complete WWPS. The control panel shall be shock-mounted to protect the instruments from breakage during transport and operation. The control panel shall have provisions for night lighting with individual switches for the lights. The control panel shall have a complete electrical schematic etched in non-corrosive metal or scaled in plastic and attached to

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the panel. The control panel shall have provisions for emergency shutdown procedures.

3.10.2.3 Hose reel assembly. The reel assembly shall include 600 feet of hard wall hose with the necessary couplings or fittings, electrical cable and support cable that meets FDA approval for potable water. The reel assembly shall be electrically powered for lowering and raising, using the onboard DEPU or an external electrical source. The reel assembly shall have provisions for the electrical cable to pass through a rotating weatherproof collector ring for supplying power to the pump assembly. The reel assembly shall have provisions for a hose guide to prevent the hose from overlapping when retrieving the hose from the well. The hose guide assembly shall have a minimum of 1,800 lbs. thrust capacity.

3.10.2.4 Submersible pump assembly. The pump assembly shall be capable of delivering 50 gallons per minute when pumping from 600 foot depths. The pump assembly shall be designed to prevent hose and electrical wire wind up during start up. The pump assembly shall be of continuous operation design and shall be capable of pumping water continuously at water temperatures of 120 °F. The pump assembly shall be multistage and fit inside a 6 inch ID casing. The pump shall deliver a minimum of 19.5 psi at the discharge hose nozzle when pumping from a depth of 1200 feet. The pumping system shall have an air purge vacuum relief system connected to the discharge plumbing.

3.10.2.5 Derrick. The derrick assembly shall have two electrical or hydraulic powered motors, one for the derrick erection and one for lowering the pump assembly into the well. The derrick assembly shall be designed for one man operation and control.

3.10.2.6 Flow meter. The WWPS shall have a flow meter showing the GPM output at the discharge hose.

3.10.2.7 Night lighting. The WWPS shall have provisions for night operation. The WWPS shall have a minimum of two shockproof quartz iodine lights or equal. The light switches shall be mounted within easy reach of the operator when standing on the ground.

3.11 Logging unit. When specified, see 6.2, a logging unit shall be furnished.

3.11.1 Description. The logging unit shall be portable and electrically operated. The electrical power supply shall be 12 volt.

3.11.2 Components. The logging unit shall include, but not be limited to, the following components:

a. Chart logging capability.
b. Electrical hoist.
c. Resistance cable head.

3.11.2.1 Chart logging. The logging unit shall have chart logging capability with minimum 10 inch chart readout capability. The unit shall have a minimum of two channels with automatic measuring system. The unit shall have

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selective chart scales to match the channel's available for varying hole depth. The unit shall be capable of performing caliper logging. The caliper as a minimum shall have three (3) arms and the ability of opening and closing from the surface while suspended in the hole.

3.11.2.2 Electrical hoist. The electrical hoist shall be of variable speed both forward and reverse. The unit shall have automatic level wind, with wire and minimum depth capability of 1500 feet.

3.11.2.3 Resistance head. The logging unit shall include capability to run down the hole spontaneous potential and single point resistance surveys simultaneously. The head assembly shall include a cable head assembly with mud pit electrodes. The head shall include Gamma Ray surveying capability without modification of the basic equipment. The unit shall have the capability of down the hole sound or a water level indicator.

3.12 Hand tools. The following tools shall be furnished with the drilling machine when required, see 6.2, to support well drilling requirements:

Item	Description	Quantity
1.	Steel wire brushes.	4
2.	Rod dope brushes.	4
3.	File, mill bastard with handle, 12 in. long.	2
4.	Hammer, sledge, double faced with handle, 10 lbs.	1
5.	Hammer, ball peen, 1-1/2 lbs.	1
6.	Knife, putty, 1-1/4 in. wide.	2
7.	Grease gun, lever hand operated.	2
8.	Oiler, hand pump type.	2
9.	Oil can, pressure type.	1
10.	Pick with handle.	1
11.	Shovel, round point.	2
12.	24 in. pipe wrenches, aluminum construction.	2
13.	36 in. pipe wrenches, aluminum construction.	2
14.	48 in. pipe wrenches, aluminum construction.	2
15.	12 in. wrench, open end adjustable.	2
16.	Allen wrenches with assortment of set screws.	1 set
17.	Jack, 10 ton capacity, hydraulic, with handle.	1
18.	Pliers, vice grip, 10 in.	2
19.	Pliers, 6 in. slip joint.	2
20.	Screwdriver, single slot type, 6 in. - 12 in.	1 set
21.	Tong, pipe, chain, adjustable for 2 in. - 12 in. O.D. pipe.	2
22.	Steel tape, 50 ft.	1
23.	Level, spirit, 24 inch, Aluminum	1
24.	Hat, hard safety type with adjustable headband (Red)	6
25.	Funnel, 12 inch diameter, galvanized, filter type	1
26.	Vice, chain, topscrew, bench, 1/4 in. - 6 in. capacity for use with pipe.	3
27.	Saw, hand, for cutting pipe w/ 48 blades, assorted.	1 set
28.	Strainer, tea, 6 in. dia. #30 mesh screen w/wood handle.	3

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29.	**Wrench set, socket, 1/2 in sq. drive, sockets 3/8 in. - 1-1/4 in.**	**1 set**
30.	**Wrenches, box combination, open end, 3/8 in. - 2 in.**	**1 set**
31.	Shovel, square point, long handle	2
32.	Circular saw, 7-1/4 in. Dia., 110 volt.	1
33.	Blades for 7-1/4 in. Dia. circular saw	
	a. Chisel tooth, combination.	2
	b. carbide tip, 40 tooth min.	4
	c. Plywood blade (flat ground).	2
	d. Rip blade.	2
	e. Cross cut.	2
	f. Metal cutting.	4
	g. Plastic cutting blade, 200 tooth	12
34.	24" Adjustable wrench	2
35.	48" Pipe wrench, Steel handle	2
36.	1 1/2" - 8" Chain tongs, Gearwrench Model C14-P	2
37.	1/4" - 2" Chain wrench, Rigid Model C-14	2

3.13 Accessories. When specified, see 6.2, the drilling machine shall be provided with the following drilling accessories or their equivalent, necessary to drill a 1500 foot water well.

Item		Quantity
1.	Spider, for setting 8 5/8" casing	1
2.	Spider, for setting 6 5/8" casing	1
3.	Spider, for setting 4" casing	1
4.	Sub, swage 2 7/8" IF box X 2" NPT Pin	1
5.	Bit, 7 7/8" 3-cone rock with 4 1/2" Reg Pin (hard formation)	1
6.	Bit, 7 7/8" 3-cone rock with 4 1/2" Reg Pin (medium hard formation)	1
7.	Casing elevator, 1" pipe	2
8.	Casing elevator, 2" pipe	2
9.	Casing elevator, 3" pipe	2
10.	Casing elevator, 4" pipe	2
11.	Casing elevator, 6" pipe	2
12.	Casing elevator, 8" pipe	2
13.	Casing elevator, 14" pipe	2
14.	Slips for 8 5/8" casing	1
15.	Swage, 8 5/8" casing X 2" NPT Box	1
16.	Casing clamp, 2 piece with 4 bolts, 1" casing	1
17.	Casing clamp, 2 piece with 4 bolts, 2" casing	1
18.	Casing clamp, 2 piece with 4 bolts, 3" casing	1
19.	Casing clamp, 2 piece with 4 bolts, 4" casing	1
20.	Casing clamp, 2 piece with 4 bolts, 6" casing	1
21.	Casing clamp, 2 piece with 4 bolts, 8" casing	1
22.	Casing clamp, 2 piece with 4 bolts, 14" casing	1
23.	Kelly wear sub, 2 7/8" IF Box X 4" IF Pin	2
24.	Bit sub 2 7/8" IF Box X 3 1/2" Reg. Box, 30" long	2
25.	Bit sub 2 7/8" IF Box X 4 1/2" Reg. Box, 30" long	2
26.	Bit sub 4" IF Box X 6 5/8" Reg. Box, 30" long	2

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27.	Crossover sub 2 7/8" IF Box X 4" IF Pin	2

Drill bits, 12 1/4 inch for 8 5/8 inch OD casing

1.	Bit, 12 1/4" 3-cone rock with 6 5/8" API Reg Pin (medium hard formations)	2
2.	Bit, 12 1/4" 3-cone rock with 6 5/8" API Reg Pin (hard formations)	2

Hughes-Hawthorne Series SH-800 bit Reamer or equal consisting of:

1.	3 blade bit shank	1
2.	4 7/8" locking bowl	1
3.	Digging reamer for 17"	1
6.	17", 3-way premium insert blades	2
7.	Swage, 14" casing X 2" NPT Box	1
8.	Slips, 14" casing	1

Accessory Equipment for Down-the-Hole-Drilling (bottom hole percussion tools will be Mission Megidril or equal):

1.	Percussion tool 5 3/8" OD X 2 7/8" IF Box, 100 - 200 psi	1
2.	Top sub, 2 7/8" IF Box threads for percussion tool	1
3.	6" Hammer bit, concave center	2
* **4.**	**7 7/8" Hammer bit, drop center**	**2**
5.	Bit breaker, 6" bit	1
6.	Bit breaker, 7 7/8" bit	1
7.	Breakout Tongs, gear wrench, 5 3/8" to 2"	1
8.	Rock Drill lube oil, Texaco or equal, 55 gal. drum	1
9.	Breakout forks for 5 5/8" hammer	1
10.	7 7/8" Over hammer reamer	1

Accessories for drilling operations:

1.	Cement Float Shoe, with check for 8 5/8" casing	2
2.	Chaser Plug for 8" ID casing	2
3.	Safety Clamp, for up to 14" pipe, with wrench and box	1
4.	Cable Sling, 1" Dia. X 6 ft. with shackle end and eye end	1
5.	Die over shot, with 2 7/8" IF Box thread (for catching 4 1/2" Dia. drill pipe)	1
6.	Taper Fishing Tap, with 2 7/8" IF Box thread for 2" ID drill pipe	1
7.	Circulating over shot, with 2 7/8" IF Box threads, for 5" minimum OD Dia. pipe	1
8.	Roller Type Drive Slips for 4 1/2" drill pipe	1
9.	Bottom hole stabilizer, 60" long with 3 straight ribs, 12 1/4" OD on 6" OD body, 6 5/8" Reg. box bored for float X 4" IF box with breakout flats	1
* **10.**	**Bottom hole stabilizer, 78" long with 3 straight ribs, 7 7/8" OD on 4 1/2" OD body, 4 1/2" Reg. box bored for**	

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	float X 2 7/8" IF box with breakout flats	1
11.	Sieve Analysis Kit, from 0.01 to 200 mesh	1
12.	# 4 Brush, for tool joint compound	4
13.	Tool Joint Compound, 2.56 gal pail	4
14.	Safety Belt, with tether (Nylon)	1
15.	Valve Seat Puller (mud pump)	1
16.	Liner Puller (mud pump)	1
17.	Wrenches, mud pump	1 set
18.	Suction Hose, 5" X 20 ft. with 5" male fittings and 5" camlock fitting	1
19.	Coupler, 5" Style 'D'	2
20.	Adapter, 5" Style 'A'	2
21.	Gaskets, 5"	2
22.	Suction Strainer with check, 5"	1
23.	Hose, High Pressure, 2" X 25 feet, 2H wire braid with 2" camlock fittings	2
24.	Hose, High Pressure, 2" X 25 feet, 2H wire braid with 2" hammer unions	2
25.	Coupler, 2" Style 'D'	2
26.	Adapter, 2" Style 'A'	2
27.	Gasket, 2"	2
28.	Hose, mixer, 1" X 25 feet, 2H wire braid with 1" camlock fittings	1
29.	Coupler, 1" Style 'D'	2
30.	Adapter, 1" Style 'A'	2
31.	Gasket, 1"	2
32.	Jet, Air Injection tool with 2 7/8" IF Box threads	1
33.	Meter, Water level, 900' capacity, with 5' increments	1
34.	Fluke, Model 73, series 11 multi-meter or equal	1
35.	Baroid, Quick-gel, 50 lb bag	160
36.	Baroid, Barafos, 50 lb bag	4
37.	Baroid, Ez-mud, 5 gal can	4
38.	Baroid, Con-dent, 5 gal can	2
39.	Baroid, Quick-foam, 5 gal can	2
40.	Baroid, Quick-trol, 21 bags per case	2 cases
41.	Baroid, Benseal, 50 lb bag	12
42.	Baroid, Soda ash, 50 lb bag	1
43.	Shale Trap, 12 1/4" hole for 8 5/5" OD casing	1
44.	Cement Basket, 12 1/4" hole for 8 5/8" OD casing	1
45.	Plastic Sheet, roll, 6 mil thick	1
46.	HTC, Calcium Hypochloride Pellets, 2.5 lb. bottle	1 case
47.	Breakout forks for 4 1/2" OD drill pipe/collar	2
48.	Breakout forks for 6" drill collars	2
49.	Check valve (float) for 6 5/8" Reg. box	2
50.	Check valve (float) for 4 1/2" Reg. box	2
51.	Check valve (float) for 3 1/2" Reg. box	2
52.	Drill pipe, light weight steel, 4 1/2" OD X 20 ft. long, 0.25 wall, grade D, Internal upset ends, inertia welded tool joints with 2 7/8" IF box threads with breakout flats 4" long and 3" across with metal thread protectors	75
53.	Drill pipe, same as above except 5 ft. lengths	2

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54.	Swivel Bail Hoisting plug, with 2 7/8" IF Pin	2
55.	Swivel Bail Hoisting plug, with 4" IF Pin	2
56.	Drill Collars, 4 1/2" OD X 2" ID X 20 ft. long with 2 7/8" IF Box to Pin threads, with breakout flats, 4" long and 3" across with metal thread protectors	4
57.	Drill Collars, 6" OD X 1/4" ID X 20 ft. long with 4" IF Box to Pin threads, with breakout flats, 4" long and 3" across with metal thread protectors	2
58.	6" Tricone combination air and mud button bit, 3 1/2" Reg. Pin	1
59.	Suction hose 2 1/2" X 25' with camlock fittings	2
60.	One cone desander with camlock fittings	2
61.	Banding tool	1
62.	Hoist plug, 2 7/8" IF Pin	2
63.	Hoist plug, 4" IF Pin	2
64.	Shackle, 3/4"	4
65.	Shackle, 1/2"	4
66.	Wash down hose, 3/8" X 50' with quick disconnect and washdown nozzle	1
67.	Filter cake paper (Box)	1
68.	CO_2 Cartridges (Box)	2

Mud Pump Repair Parts

1.	Valve Seat	16
2.	Valve Springs, Brass	16
3.	Valve Assembly	16
4.	Valve Inserts	16
5.	Liners	4
6.	Gasket, Liner	8
7.	Spacer, Liner	4
8.	Clamp, Liner	4
9.	Piston Rod	2
10.	Piston Assembly (kit)	4
11.	Gasket, Head	4
12.	Packing, Rod	24
13.	Gasket, Valve	16
14.	Packing, OSH	6
15.	Packing Ring	36
16.	Gasket, SF	6
17.	Liner Sleeve	8

3.14 Well Completion Kit. When specified, see 6.2, a well completion kit shall be furnished.

3.14.1 Well Completion Kit. The following items shall be included with the well drilling system. Each item shall have complete instructions on utilization and operation when applicable.

Item	Description	Quantity
1.	Well Casing, PVC, 14 in. OD X 1/2 in. wall, 20	6

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	feet long.	
2.	Couplings, for 14 in. OD PVC casing.	6
3.	Accessories for cutting, regrooving and locking (pin driver), for 14 in. PVC Certalock or equal.	1 set
4.	Well Casing, seamless, 8-5/8 in. OD X 332 wall, 12.79 per ft., API SL grade 'B', black threaded and coupled with thread protectors, 20 ft. long, wt lbs/ft 28.55	60
5.	Well screen, 8 in. (#304 stainless steel) extra heavy, 015 slot, 10 ft. lengths male and female fittings with thread protectors.	6

3.15 Spare Parts Kit. The supplier of the well drilling machine, well drilling support vehicle, well drilling fluid handling system, water well pumping system, and the logging unit shall furnish a spare parts kit that will ensure operational capability for a minimum period of 300 hours operation. A list of the parts furnished in the parts kits shall be submitted for review at the time of bid submission for completeness, cost and approval.

3.16 Lubrication Plate. A lubrication plate showing the following information shall be securely attached to each well drilling machine and support vehicle:

a. Point of application.
b. Service interval.
c. Type of lubrication.
d. Viscosity.

3.17 Special Tools. Any special tools required for adjustments to support the system's operation including special category tools to meet air transportability requirements, slings, jigs, sockets, socket wrenches larger than 1-1/4 inch, shall be furnished by the contractor.

3.18 Toolbox. When special tools are provided, see 3.15, a closed compartment or toolbox shall be provided to carry them on the well drilling machine. Toolbox or compartment shall be made of sheet steel of not less than 0.0747 inch nominal thickness (U.S. standard gauge #14.). The toolbox shall have a hinged lid and means that will keep the lid closed when the toolbox is subjected to vibration. The toolbox shall be securely fastened to the unit in a protected and accessible location.

3.19 Sound Level. When measured in accordance with 4.5.2, the sound level at the operators station shall not exceed 85 dBa as shown in MIL-STD-1474. In the event of failure to meet this requirement the contractor shall provide hazard warning plates or decals which conform to MIL-STD-1474.

3.20 Air Transportability. The truck mounted well drilling machine and support vehicle shall be air transportable without disassembly in a drive-on/drive-off mode from ground level to aircraft and vice versa. All truck or trailer mounted components of the system shall meet or exceed all specified requirements in MIL-STD-1791 applicable to transport on C-130, C-141, C-17 and C-5 aircraft, (see 6.2 item n).

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3.20.1 Removal of Items for Air Transportability. Components, such as the mast, cab, mud pump, hoist assembly, drive train, etc., may not be removed **either whole or partially from the drilling machine to meet dimensional or** weight requirements for aircraft drive-on loading. Using a mechanic with common hand tools, or any special tools required, the crane mounted on the support vehicle and the operator who assists the mechanic, neither the removal nor the replacement of such parts shall require longer than 2 hours. No accessories or attachments which would prevent the drilling machine from being driven on/off the aircraft shall be removed. The operator's manual shall include complete instructions for both component removal and replacement. Items that may be removed are personnel platforms, maintenance stands, catwalks, guides and similar parts and assemblies. The mast shall not be used as a removable component for meeting the weight requirements specified herein. The weight of individual parts or assemblies which must be handled or positioned shall not exceed the limits specified in MIL-STD-1472 for a two person lift. All necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. If the item or assembly is removable it shall be capable of being removed without special tools. Fasteners securing the items shall not be subject to loosening by vibrations experienced during operation of the vehicle.

3.20.2 Removal of Items for Air Transportability Video Tape. The contractor shall provide a VHS format video tape consisting of step-by-step instructions for preparing the well drilling unit for the requirements of para. 3.20. The video tape shall be in natural color showing all required lift/balance points, slings, points of component removal, all safety requirements, reassembly and all required adjustments to begin drilling operations.

3.21 Cleaning, painting plating anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with FED-STD-297, except the commercial components may be prepared in accordance with the manufacturer's standard practice provided it is compatible with exterior finish color.

3.21.1 Paint. The outer surface of the well drilling system and the interior compartments (except the cab) shall be painted using the manufacturer's standard lusterless commercial grade paint conforming to Color Number 24052, Forest Green or Color Number 30313, Desert Sand IAW FED-STD-595. All exterior trim items shall be painted with the same paint as the exterior of the vehicle. Interior upholstery shall be dark in color and of extreme heavy duty service material. The manufacturer's practices and painting techniques shall comply with MIL-STD-1223. Special attention shall be given to utilizing lead and chromate free paints. The vehicle shall be rustproofed and undercoated in accordance with FED-STD-297D, para 5.8 for tropical rustproofing. The selected paint shall have a minimum pencil hardness of 6H when tested IAW paragraph 4.8.12 of MIL-C-83286, demonstrate resistance to taping IAW paragraph 3.7.3.7 of MIL-C-83286 and be capable of passing the adhesion test as outlined in 4.4.8 of this specification.

3.21.2 Markings. Unless otherwise specified, markings for vehicles painted Forest Green shall be applied with the manufacturer's compatible standard grade

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lusterless black paint conforming to Color Number 37038, and markings for vehicles painted Desert Sand shall be lusterless white conforming to Color Number 37875 IAW FED-STD-595.

3.22 Identification Marking. A corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each machine in the cab.

a. Nomenclature.
b. Manufacturer's model number.
c. Manufacturer's serial number.
d. Special characteristics, HP, RPM, etc.
e. Manufacturer's name.
f. Contract number.
g. Date of manufacture.
h. Weight.
i. Shipping cube and center of gravity.
j. U.S.

3.23 Workmanship.

3.23.1 Steel Fabrication. Steel used in the fabrication of equipment shall be free of kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and shaping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.23.2 Fasteners. Ferrous alloy fasteners shall be cadmium plated per **QQ-P-416 Type II, to a minimum thickness of 0.0003 inches unless threaded** tolerances dictate a different thickness and aluminum alloy fasteners shall be anodized per MIL-A-8625. Fastener installation shall be IAW MIL-STD-1568. Fastener quality shall be IAW US Air Force Drawing 8838302A.

3.23.2.1 Bolted Connections. Bolt holes shall be accurately punched or drilled with all burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice and all bolts, nuts and screws shall be tight.

3.23.2.1.1 Fastening devices. All screws, bolts, nuts, and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Lockwashers, nylon patches or cotter pins shall not be used for this purpose, except for attachment of trim items or as provided in commercial (see 6.4.2) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter. All fasteners shall follow the quality assurance requirements of Air Force Drawing 8838302A.

3.23.2.2 Riveted Connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivets. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the

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member.

3.23.2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.23.3 Welding. The surface of parts to be welded shall be free from weld slag, rust, scale, paint grease or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall conform to the procedures as prescribed by either the AWS or ASME for the type of welding operation to be performed.

3.23.4 Machine Work. Tolerances and gauges for metal fits shall conform to the limitation specified herein and on the applicable drawings, and otherwise to the standards of good commercial practice.

3.23.5 Castings. Castings shall be sound and free from defects and voids.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of testing. Testing shall be classified as follows:

a. First article testing (see 4.2.1).
b. Quality conformance testing (see 4.2.2).
c. User test (see 4.4.7).

4.2.1 First article test sample. The first article test sample may be a preproduction item or production item which meets the requirements of this specification. The sample shall be identical to the items procured under the

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terms of the contract. Approval of the first article sample shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.2.1.1 Test plan. The test sample shall be tested as follows:

a. Phase I. The well drilling system shall be subjected to the examination of 4.3 and the tests described in 4.4 through 4.5, except 4.4.4 and 4.4.7, at a location chosen by the contractor and approved by the Government. Testing shall not begin without minimum 30 day prior notification to the contracting **officer and WR-ALC/LVRS personnel.**

b. Phase II. Upon successful completion of Phase I, the well drilling system shall be moved to the field testing location as prescribed by the contracting officer for testing as described in 4.4.4. The performance test shall not begin until after completion of Phase I of first article testing as defined by 4.2.1.1.a. The vehicle shall not be moved to the field testing site without prior 15 day notification to the PCO. Testing shall begin within 2 weeks after completion of Phase I testing and shall conform to all local and state laws as applicable to the location of the test site. The preliminary technical data shall contain procedures for the preparation and tiedown of the vehicle for shipment. The contractor shall provide any special tools and equipment required to perform operators maintenance and scheduled maintenance. Any parts or corrective actions required shall be provided by the contractor. Any maintenance actions performed by the contractor shall not begin before notification to a field representative, and government personnel shall be allowed to observe their actions. The contractor shall provide all vehicle consumables required in the course of the test except as indicated by the contracting officer and the prescribed test facilities. All liabilities shall be the responsibility of the contractor during testing unless otherwise indicated by the contracting officer and the testing field representative(s). Noncompliance with the performance **requirements of this document shall be cause for rejection. The contractor** shall maintain a daily log. The log shall be signed by the test site field representative and the contractor after each day of activity. The daily log shall document, but not be limited to the following:

a. Operating time.
b. Tasks completed.
c. Technical manual errors.
d. All failures and corrective actions taken.
e. Operational problems encountered.
f. Test site field observer comments.

4.2.2 Quality conformance testing. Quality conformance testing shall be classified as follows:

a. Individual testing (see 4.2.2.1).
b. Random sample testing (see 4.2.2.2).

4.2.2.1 Individual testing. Each well drilling system or individual end item of the system shall be subject to the test specified in 4.5.

4.2.2.2 Random sample testing. Any production end item may be selected by

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the procuring activity at any time to be subjected to any or all of the tests described in 4.5. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets in the report.

4.2.3 Test conditions.

4.2.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.2.3.2 Data. During the testing specified herein, at least the following data, unless not applicable shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test:

a. Date.
b. Time started.
c. Time finished.
d. Ambient temperature.
e. Engine speed.
f. Engine oil pressure.
g. Engine coolant temperature.
h. Transmission oil temperature.
i. Vehicle odometer reading at the start and finish of each test.
j. Engine hour meter reading at the start and finish of each test.
k. Hydraulic pressure at the pump outlet.
l. Hydraulic fluid temperature in the reservoir.

4.2.3.3 Preparation for testing. The well drilling system and all associated support equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.2.3.4 Test rejection criteria. Throughout all tests specified, the well drilling system shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubrication or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground and all required obstacles, with the exception of normal

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contact with the tires.

f. Misalignment of components.

g. Evidence of undesirable road handling characteristics including instability in handling during cornering, braking or while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

l. Overheating of the engine, transmission or any vehicle component.

4.2.4 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report (see 6.2). The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other event recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made, and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements, that if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used in accordance with FED-STD-297.

4.3. Examination. Each water well drilling system component shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection.

4.4 Tests.

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4.4.1 Run in Test. The drilling system components produced in fulfillment of the contract shall be completely assembled, adjusted, lubricated and otherwise serviced for operation. A run-in test shall be conducted in accordance with the manufacturer's recommendations. The manufacturer may devise a separate series of tests, other than those included in Section 4, that may be used to demonstrate advantages of the equipment peculiar to the manufacturers design. All additional tests performed shall be done at the manufacturer's own expense. The manufacturer shall be solely responsible for reviewing and redesigning the system until the problem has been identified and corrected if the vehicle fails to perform any of the manufacturer's suggested tests. During this operational test all controls, adjusting mechanisms, brakes, hydraulic systems and other accessories shall be operated sufficiently to insure against restriction or malfunctions. Failure to pass any phase of this test, except those tests performed at the manufacturers own expense, shall be cause for rejection.

4.4.2 Sound Level Measurement. With the microphone located 6 inches from the operator's ear with the operator on the drill platform by the controls, measure the maximum steady state noise levels in accordance with MIL-STD-1474 for all drilling modes including full operation of all necessary support and accessory equipment. Any reading in excess of 85 decibels shall invoke the requirement for hazard warning plates, or decals, as specified in 3.19. All other components of the well drilling system will meet the requirements of MIL-STD-1474 and will have hazard warning plates when required.

4.4.3 Mobility Tests. The component(s) shall be tested with full payloads on courses described below:

a. 550 miles over primary roads, maintaining speeds between 45 and 55 mph. Speeds of 55 mph on level primary roads shall be demonstrated.

b. 300 miles over improved secondary roads, maintaining speeds between 30 and 50 mph.

c. 150 miles over cross-country including a minimum of 25 miles on a course similar to Belgian Block (see 6.4), maintaining speeds of 8 to 10 mph.

d. Service brakes shall be applied normally during the road test.

e. All lights, horns, speedometer, hourmeter, controls and instruments pertaining to the vehicle engine shall be tested and observed.

After completion of the tests specified herein, the vehicle(s) shall be checked at all locations where a permanent deformation may occur to include but not be limited to: Where components are attached to the body/frame, e.g.; engine, tanks, axles, shock absorbers, springs. Welds shall be checked with xyglo or magnaflux as applicable. All check points shall be logged, measurements and defects noted and reported in conformance to MIL-STD-831. Permanent deformation, overheating, malfunction, failure to meet performance requirements, failure to complete any test, leakage of fuel, or leakage of hydraulic oil or lubricants shall be cause for rejection of the vehicle(s).

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If such deficiencies do arise, they shall be corrected and the affected test shall be repeated as required.

4.4.4 Performance Test. All requirements of section 4.4.4 shall be performed at the test site as prescribed by the contracting officer. The well drilling system shall drill to a depth of 1,500 feet using rotary mud, rotary air, and down-the-hole hammer under the following conditions:

a. The drilling operation shall commence in full daylight and shall not exceed 160 consecutive hours including set-up and tear-down time. A minimum of four - (8) hour shifts shall include night operation using lighting from the drilling machine and support vehicle.

b. The manufacturer shall drill a 12-1/4 inch hole to 500 feet using rotary mud techniques or until reaching hard consolidated formations and continue to a depth of 1,450 feet using rotary air and down-the-hole techniques. Rotary mud shall be used to complete the well to a depth of 1,500 feet. A pilot hole to 50 feet is acceptable.

c. Drill a 12¼ inch diameter hole and set a 10 inch OD surface casing to a minimum depth of 60 feet.

d. Maintain a drilling log and indicate the following:

Formations encountered and at what depth
Drilling time
Shutdowns and reasons (including service and maintenance)

Formation samples shall be taken at 20 foot intervals and verification of formations shall be recorded in the drilling log as part of the test.

e. Install the submersible pump, drop hose, and demonstrate that all well completion material is fully functional to a minimum depth of 600 feet. The contractor shall start and run the submersible pump for not less than 1 hour.

f. The support vehicle shall fully demonstrate its capability for carrying all accessories and supplies required in support of drilling to a depth of 1,500 feet, and for providing the hoisting, winching, water/fuel pumping, welding, generating, and tool storage needs of field drilling operations.

g. Submit to the contracting officer a certified reproducible copy of the drilling log. Inability to complete the test, drilling log incomplete, or any permanent deformation or breakage of components of the drilling system shall constitute failure of this test.

4.4.5 Air Transportability Test. To demonstrate the air transportability of the well drilling machine and other components of the well drilling system, in the transportation configuration, an aircraft mockup shall be furnished by the manufacturer. The mockup shall consist of a 15 feet long 15 degree ramp connecting two horizontal ramps. The ramps should be at least 105 inches

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wide. Two rectangles, each measuring 119 inches wide and 108 inches high inside, shall be installed 90 degrees to the horizontal, one rectangle at each end of the upper ramp. This ramp represents the floor of the aircraft and the insides of the rectangles represent the inside dimensional limits of the aircraft. The longitudinal centerline of the upper ramp shall coincide with the centerline of the widths of the two rectangles. The cross section of the upper horizontal surface representing the aircraft floor shall be configured to represent the rail restriction in C-130 A/C as per MIL-STD-1791. This restricts the width to 105 inches for the first 5-1/2 inches above the floor level. The drilling system components, in the drive on/off configuration, shall then be driven or towed from ground level the full length of the mockup, and then driven or towed back to ground level again. At the supplier's option, the items may be backed into the mockup or driven forward into the mockup. Any interference between the bottom of the items and the ramps or between any parts and the simulated inside of the aircraft shall be cause for rejection. At the manufacturer's option and availability of an actual C-130 aircraft, the manufacturer may demonstrate the systems air transportability using a military version C-130.

4.4.6 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.2.2 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from the approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of performing maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.4.7 User test. When a user test is required, (See 6.2), the test vehicles shall be delivered to the prescribed test site for 45 available days (seven days a week) of user tests. The 45 available days shall be consecutive, except when interrupted for maintenance/repair downtime as defined herein. The user test will not begin until after completion of the first phase of first article testing as defined by 4.2.1.1.a. The test sample shall not be shipped to the prescribed test site without prior 15 day notification to the PCO. The test shall begin within two weeks after completion of phase I testing and shall end after 45 days of availability are completed or 60 calendar days have elapsed, whichever occurs first. If the user test sample has not been available for 45 test days after 60 calendar days have elapsed, the user test shall be considered a failure. The vehicle shall be available for any one period of eight consecutive hours during each day of the 45 day test. If 24 hour operations are scheduled, each 24 hour period shall be equal to three days of operation. The periods shall be scheduled by the test site field representative, and the schedules shall be provided to the contractor at least 24 hours in advance. Any period of more than four hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day for

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the test period. The vehicle(s) shall be operated, and scheduled maintenance shall be performed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the test sample. The preliminary technical data shall contain procedures for preparation and tiedown of the vehicle for shipment. The contractor shall provide any special tools and equipment required to perform both operators maintenance and scheduled maintenance. Any parts or corrective actions required shall be provided by the contractor. Any maintenance actions performed by the contractor shall not begin before prior notification to a field representative, and Government personnel shall be allowed to observe their actions. The contractor shall provide all vehicle consumables required in the course or the test except as indicated by the contracting officer and the prescribed test facilities. The contractor shall provide on-site familiarization to the operators and mechanics as necessary to perform this test. Test operations will be used in, or simulate normal Air Force usage and will not exceed any performance **requirements defined in Section 3 of this specification. Noncompliance with the performance requirements of this specification shall be cause for rejection.** The contractor shall maintain a daily log. The log shall be signed by the test site field representative and the contractor after each day of activity. The daily log shall document, but not be limited to the following:

a. Operating time.
b. Tasks completed.
c. Technical manual errors.
d. All failures and corrective actions taken.
e. operational problems encountered.
f. Test site field observer comments.

4.4.8 Adhesion (wet) tape test. This method covers a procedure suitable for establishing acceptability of the undercoat and surface adhesion of the paint coating system. The test will be accomplished on each production item, (e.g., drilling rig, support vehicle, air compressor and well pumping system.)

a. The test is not applicable to unit parts where the surface area or contour of the unit is such that the test would be physically impossible to accomplish, e.g., individual components such as motors or engines.

b. Adhesion tests shall be made on the completed exterior finish of the painted area after drying for a period of at least 48 hours. The test will be accomplished on only one area unless failure occurs, then several areas will be checked to determine that a satisfactory level of adhesion has been achieved. Three consecutive failures are grounds for rejection. Failures are defined as any visible peeling, chipping or defects.

c. Perform the wet tape test as follows:

1. Soak a clean cloth in distilled water (size at least 6 inches by 3 inches).

2. Fold the cloth into a square while it is still wet and place it on the area to be tested. Do not wring out the cloth.

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3. Place a piece of plastic over the wet cloth. The plastic must completely cover the cloth and hold the cloth in contact with the test surface.

4. Tape the plastic sheet onto the structure. Completely seal the edges with tape so that the water cannot evaporate.

5. Allow the wet cloth to continuously contact the area for a minimum of 24 hours.

6. Remove the wet cloth and blot up the surface water. Immediately apply a one-inch strip of tape, (Code No. 250 Minnesota Mining and Manufacturing Company, age of tape cannot exceed one year prior to date of test), adhesive side down. Press the tape down, using two passes of a 4-1/2 pound rubber covered roller or employ firm pressure with the hand for a minimum of 3 seconds.

7. Remove the tape in one abrupt motion, pulling perpendicular from the surface, and examine the tested area for any paint damage such as removal of any paint at one of the layers of the finish system or removal of the entire system from metal.

4.5 Production Unit Operation Test. Each production well drilling system component shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warmup period as recommended by the manufacturer. The unit shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction.

5. PREPARATION FOR DELIVERY AND MARKING

5.1 Vehicle processing. The well drilling system or any one part of the system delivered under this specification shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

5.2 Preservation Packaging. The manufacturer's standard commercial practice shall apply for U.S. surface shipping or overseas surface shipping of vehicles. All materials supplied by this contract shall be packed in weatherproof reusable containers. These containers shall be constructed of grade # 2 wood, or better and lined with 3/8 inch marine plywood. Containers shall have lid hasps; lid chains; dividers to secure materials; and data plates secured to the upper left, front corner of each container identifying its contents. Containers will be of the size to load on 463L pallets. Drill pipe, well casing, drill collars, and the well completion kit shall be packaged in like bundles and waterproof containers conforming to above requirements for air transportable shipment. Packaging for drill pipe, and well casing shall provide for resistance of

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individual pipes from sliding out of the center, sides, or top of the bundle. The packaging shall be reusable and undamaged due to repeated use.

5.3 Marking. The well drilling system components and attachments, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The well drilling system is to be used to drill wells for water to 1,500 feet depths in varied soil and geological conditions.

6.2 Ordering Data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Issue of DoDiss to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Well drilling system component being purchased.
(1) Well drilling machine (see 3.8).
(2) Support vehicle (see 3.9).
(3) Water well pumping system (see 3.10).
(4) Logging unit system (see 3.11).
(5) Trailer mounted air compressor (see 3.9.3).

d. When first article test is required for inspection and approval of the:

(1) Drilling machine, if required (see 3.6).
(2) Support vehicle, if required (see 3.7).
(3) Water well pumping system, if required (see 3.10).
(4) Logging unit, if required (see 3.11).
(5) Trailer mounted air compressor (see 3.9.3).

e. When user test is required for inspection and approval of the:

(1) Drilling machine, if required (see 3.6).
(2) Support vehicle, if required (see 3.7).
(3) Water well pumping system, if required (see 3.10).
(4) Logging unit, if required (see 3.11).
(5) Trailer mounted air compressor (see 3.9.3).

f. First article test report (see 4.2.4).
g. Accessories required (see 3.13).
h. Support vehicle tools required (see 3.9.2.14).
i. Hand tools required (see 3.12).
j. Well completion kit (see 3.14).
k. Color of finish coat required (see 3.22).

6.2.1 Well Drilling and Support Vehicle Chassis Requirements.

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(see para. 3.1 and 3.3). The following items from Federal specification - KKK-T-2111 (para 6.2) are required as indicated below for the well drilling unit and support vehicle:

a. Trucks and Truck Tractors; Commercial, Diesel Engine Driven, 36,000 to 60,000 Pounds GVW, 6X6, KKK-T-2111K, dated August 1, 1989.
b. Type 1 chassis, Class B.
c. See 2.1.1 and 2.2.
d. See 3.22.1.
e. See 3.22.1 and 3.22.2.
f. N/A
g. See 3.22.1.
h. N/A
i. Required, see 3.22.1.
j. Required.
k. Required for front and rear.
l. Required.
m. Required.
n. Required.
o. Required.
p. Maximum of 13,000 lb front axle and 32,000 lb live tandem rear axles are required. Suspension necessary to meet GVW is required.
q. N/A
r. 46,000 minimum required.
s. Required.
t. Required.
u. Engine manufacturer's recommended filter.
v. Not required.
w. Not required.
x. Not required.
y. Not required.
z. Minimum alternator capacity of 85 amperes is required.
aa. Compatible with water well pumping system trailer and air compressor trailer.
ab. Not required.
ac. Dry type, two-stage air cleaner with dash mounted service indicator required.
ad. As specified in 3.4.3.2 except fuel capacity shall allow a minimum of 8 continuous hours of operation before refueling.
ae. Required.
af. Not required with turbocharged engine.
ag. Required.
ah. Transmission provided shall be a manual type complying with all applicable paragraphs IAW 3.4.5 of specification KKK-T-2111.
ai. Heavy duty frame 2.8 million minimum, Resisting Bending Moment (RBM) for the well drilling unit and 2.5 million minimum RBM for the support unit.
aj. Required.
ak. Required.
al. See 3.5.1.
am. Single front and dual rear bogies, total 10 wheels, 10 stud disc type wheels.

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an. See 3.5.1.
ao. Required.
ap. Not required.
aq. Required.
ar. Required.
as. Not required.
at. Manufacturer's standard.
au. Required, with seat belts.
av. Required, both seats.
aw. Not required.
ax. Required, 20 ton jack, handle and wheel nut wrench for support vehicle. A wheel nut wrench is required for the well drilling rig.
ay. Required.
az. Required.
ba. Required.
bb. Required.
bc. Required.
bd. Not required.
be. See 3.5.
bf. See 3.5.2.
bg. Manufacturer's standard dimensions meeting all air transportability requirements (see 3.20).
bh-cb. N/A
cc. Required, two copies of parts list and maintenance/repair manuals shall be provided for each vehicle.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential are requested or provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	Data Requirement Title	Applicable DID Number
4.2.1.1(a and b)	Notification of Tests	DI-T-2373A/T
4.2.3.4, 4.2.4, 4.4	First Article Inspection	DI-T-4902

6.4 Preproduction Sample. The sample (First Article) may be a production **item or a preproduction item which meets the requirements of this specification.** In either case, the sample shall be identical with the proposed

item and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a preproduction sample shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification. The preproduction sample may be shipped as part of

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the contract.

6.5 Definitions.

6.5.1 Belgian Block. A Belgian Block course is paved with unevenly laid granite or basalt block forming an undulating surface. It duplicates a rough cobblestone road such as is found in many parts of the world. A cobblestone course may be used as an alternate.

6.5.2 Commercial. A manufacturer's standard (product) having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes or experimental units do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales (see 3.4).

6.6 Subject term (key word) listing.

Air hammer
Air Transportable, C-5
Air Transportable, C-17
Air Transportable, C-130
Air Transportable, C-141
Baroid
Commercial
Crane
Crown Block
Derrick
Diesel
Drawworks
Engine, diesel
Hydraulic
JP-4
JP-5
JP-8
Mast
Mud
Pump, mud
Rock
Sand
Soil
System, pumping
Truck
Vehicle, Support
Vehicle
Waterbed
Winch

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